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Indexing System for Milling Tools

Abstract

An example indexing system has a tool body, a cam, a carrier rod, and an indexable cutting insert. The tool body is rotatable around a tool body central axis and has a bore. The cam is attached to the tool body and is rotatable around the tool body central axis between a cam first position and a cam second position. The carrier rod is partially disposed within the bore, contacts the cam, and is moveable between a carrier rod first position when the cam is in the cam first position and a carrier rod second position when the cam is in the cam second position. The indexable cutting insert is attached to the carrier rod and is moveable between a clamped position when the carrier rod is in the carrier rod first position and an indexing position when the carrier rod is in the carrier rod second position.

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Background/Summary

FIELD

[0001] The disclosure relates generally to the field of milling tools. More particularly, the disclosure relates to an indexing system for milling tools.

BACKGROUND

[0002] Milling tools are generally equipped with a relatively large number of indexable cutting inserts that require indexing when the cutting edges become worn. Indexing is usually performed manually such that each cutting insert is detached, indexed with an unused cutting edge arranged for active cutting, and then mounted again. Manually indexing cutting inserts is time-consuming and costly as it requires the milling tool to be taken out of operation during indexing.

[0003] Prior attempts at performing indexing operations in milling tools have incorporated, among other components, springs to move the inserts between clamped and indexing positions. However, the use of springs increases the overall cost of manufacturing a device and reduces the stiffness and clamping force used to hold an insert in place, which can lead to decreased performance and repeatability issues. In addition, using springs increases the complexity associated with assembling and maintaining the device.

[0004] A need exists, therefore, for new and useful indexing systems for milling tools.

SUMMARY

[0005] Various example indexing systems for milling tools are described.

[0006] An example indexing system has a tool body, a cam, a carrier rod, and an indexable cutting insert. The tool body is rotatable around a tool body central axis. The tool body has a tool body central passageway, a bore, and an insert pocket having a plurality of side contact surfaces. The cam is attached to the tool body, is disposed within the tool body central passageway, and is rotatable around the tool body central axis between a cam first position and a cam second position. The carrier rod is partially disposed within the bore, contacts the cam, and is moveable between a carrier rod first position and a carrier rod second position. The carrier rod is in the carrier rod first position when the cam is in the cam first position. The carrier rod is in the carrier rod second position when the cam is in the cam second position. The indexable cutting insert is attached to the carrier rod and has an insert center axis and a plurality of side support surfaces. The indexable cutting insert is moveable between a clamped position and an indexing position. The indexable cutting insert is in the clamped position when the carrier rod is in the carrier rod first position. The indexable cutting insert is in the indexing position when the carrier rod is in the carrier rod second position. In the clamped position, a first side support surface of the plurality of side support surfaces contacts a first side contact surface of the plurality of side contact surfaces and a second side support surface of the plurality of side support surfaces contacts a second side contact surface of the plurality of side contact surfaces. In the indexing position, the first side support surface does not contact the first side contact surface, the second side support surface does not contact the second side contact surface, and the indexable cutting insert is rotatable around the insert center axis.

[0007] Additional understanding of these examples can be obtained by review of the detailed description, below, and the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a first embodiment of an indexing system. The indexable cutting inserts are illustrated in a clamped position.

[0009] FIG. 2 is another perspective view of the indexing system shown in FIG. 1. The indexable cutting inserts are illustrated in an indexing position.

[0010] FIG. 3 is an exploded view of the indexing system shown in FIG. 1.

[0011] FIG. 4 is a top view of the indexing system shown in FIG. 1. The indexable cutting inserts are illustrated in a clamped position.

[0012] FIG. 4A is a magnified view of area 4A shown in FIG. 4.

[0013] FIG. 5 is another top view of the indexing system shown in FIG. 1. The indexable cutting inserts are illustrated in an indexing position.

[0014] FIG. 6 is a magnified view of a first side of an indexable cutting insert shown in FIG. 1.

[0015] FIG. 7 is a magnified view of a first side of an indexable cutting insert shown in FIG. 2.

[0016] FIG. 8 is a magnified view of a second side of an indexable cutting insert shown in FIG. 1.

[0017] FIG. 9 is a magnified view of a second side of an indexable cutting insert shown in FIG. 2.

[0018] FIG. 10 is a partial perspective view of the indexing system shown in FIG. 1.

[0019] FIG. 11 is a perspective view of a carrier rod of the indexing system shown in FIG. 1.

[0020] FIG. 12 is a perspective view of a shim of the indexing system shown in FIG. 1.

[0021] FIG. 13 is a sectional view of the indexing system shown in FIG. 1.

[0022] FIG. 14 is a perspective view of a second embodiment of an indexing system. The indexable cutting inserts are illustrated in a clamped position.

[0023] FIG. 15 is a sectional view of the indexing system shown in FIG. 14.

[0024] FIG. 16 is a perspective view of a carrier rod of the indexing system shown in FIG. 14.

[0025] FIG. 17 is a perspective view of an alternative carrier rod shaft that can be included in an indexing system.

DETAILED DESCRIPTION

[0026] The following detailed description and the appended drawings describe and illustrate various example embodiments of an indexing system. The description and illustration of these examples are provided to enable one skilled in the art to make and use an indexing system according to this invention. They are not intended to limit the scope of the claims in any manner.

[0027] FIGS. 1 through 13 illustrate a first example indexing system 10. The indexing system 10 has a tool body 12, a cam 14, a plurality of carrier rods 16, a plurality of indexable cutting inserts 18, a plurality of shims 20, and a retaining ring 22.

[0028] As shown in FIGS. 1 through 10 and 13, the tool body 12 is rotatable around a tool body central axis 13 and has a tool body first end 24, a tool body second end 26, a tool body periphery 28, a tool body central passageway 30, a tool body recess 32, a tool body first shoulder 34, a tool body second shoulder 36, a plurality of bores 38, a plurality of slots 40, and a plurality of insert pockets 42.

[0029] The tool body central passageway 30 extends through the tool body 12 and from the tool body first end 24 to the tool body second end 26. The tool body recess 32 extends from the tool body central passageway 30 into the tool body 12, around the tool body central axis 13, and is sized and configured to receive a portion of the retaining ring 22, as described in more detail herein. The tool body first shoulder 34 is disposed between the tool body recess 32 and the tool body second shoulder 36. The tool body first shoulder 34 provides a mechanical stop to advancement of the cam 14 during assembly. The tool body second shoulder 36 is disposed between the tool body first shoulder 34 and the tool body first end 24 and provides a mechanical stop to advancement of a locking bolt 44, as shown in FIG. 13 and described in more detail herein.

[0030] As shown in FIGS. 3 and 10, each bore of the plurality of bores 38 has a bore lengthwise axis 45, a bore first end 46, and a bore second end 48. Each bore of the plurality of bores 38 extends from a respective bore first end 46, which is disposed at the tool body central passageway 30, toward the tool body periphery 28 to a respective bore second end 48. The bore second end 48 is disposed at the slot 40. Each bore of the plurality of bores 38 extends from the bore first end 46 to the bore second end 48 and is sized and configured to receive a portion of a respective carrier rod of the plurality of carrier rods 16. Each slot of the plurality of slots 40 extends from the tool body periphery 28 toward the tool body central passageway 30 to a respective bore of the plurality

of bores **38** and is sized and configured to receive a portion of a respective carrier rod of the plurality of carrier rods **16**.

[0031] As shown in FIGS. **6**, **7**, **8**, and **9**, each insert pocket of the plurality of insert pockets **42** has an insert pocket terminal end **50**, an insert pocket tangential wall **52**, and a plurality of side contact surfaces **54**. Each slot of the plurality of slots **40** extends from a respective insert pocket terminal end **50** toward the tool body central passageway **30** and into the insert pocket tangential wall **52**. In the embodiment shown, a bore of the plurality of bores **38** extends toward a respective insert pocket tangential wall **52** from the bore first end **46** to the bore second end **48**. The bore lengthwise axis **45** of each bore of the plurality of bores **38** is not parallel to a respective the insert pocket tangential wall **52**. As shown in FIG. **10**, the bore lengthwise axis **45** of each bore of the plurality of bores **38** is disposed at an angle **51** relative to a hypothetical plane **53** that is disposed parallel to a respective insert pocket tangential wall **52**. This structural arrangement provides a mechanism for pulling an indexable cutting insert toward an insert pocket tangential wall when moving from an indexing position to a clamped position and for pushing the indexable cutting insert away from the insert pocket tangential wall when moving from a clamped position to an indexing position. In the illustrated embodiment, the angle **51** is equal to about 5 degrees. However, in alternative embodiments a bore lengthwise axis of each bore of a plurality of bores can be disposed at any suitable angle relative to a hypothetical plane that is disposed parallel to a respective insert pocket tangential wall. Examples of angles considered suitable include angles equal to, less than, greater than, or about 1 degree, 2 degrees, 3 degrees, 4 degrees, 5 degrees, 6 degrees, 7 degrees, 8 degrees, 9 degrees, 10 degrees, and any other angle considered suitable for a particular embodiment.

[0032] While the tool body **12** has been illustrated as having a particular structural arrangement and as including a tool body recess **32**, a tool body first shoulder **34**, a tool body second shoulder **36**, a plurality of bores **38**, a plurality of slots **40**, and a plurality of insert pockets **42**, a tool body can include any suitable structural arrangement and number of features. Selection of a suitable structural arrangement for a tool body and of a number of features to include on a tool body can be based on various considerations, including the intended use of the indexing system of which the tool body is a component. Examples of numbers of bores, slots, and insert pockets considered suitable to include on a tool body include one, at least one, two, a plurality, three, four, five, six, seven, eight, nine, ten, more than ten, and any other number considered suitable for a particular embodiment. In the illustrated embodiment, the plurality of bores **38** includes ten bores, the plurality of slots **40** includes ten slots, and the plurality of insert pockets **42** includes ten insert pockets.

[0033] The cam **14** is attached to the tool body **12**, is disposed within the tool body central passageway **30**, and is rotatable around the tool body central axis **13** between a cam first position, as shown in FIGS. **1**, **4**, and **13**, and a cam second position, as shown in FIGS. **2** and **5**. In the embodiment shown, the cam **14** is releasably attached to the tool body **12** and has about 36 degrees of rotation around the tool body central axis **13**, which results in complete movement of each carrier rod of the plurality of carrier rods **16** through a respective cam first slot **62**, as described herein. However, alternative embodiments can include a cam that has any suitable degree of rotation around a tool body central axis, such as those greater than, or less than, 36 degrees.

[0034] In the illustrated embodiment, the cam **14** is rotatable around a cam central axis **55** and has a cam first side **56**, a cam second side **58**, a cam periphery **60**, a plurality of cam first slots **62**, a plurality of cam second slots **64**, and a cam passageway **66** that extends from the cam first side **56** to the cam second side **58**. In the embodiment shown, the cam central axis **55** is coaxial with the tool body central axis **13**. A cam can rotate between a cam first position and a cam second position by inserting a tool within a cam passageway, such as a hexagonal tool, and rotating the cam counterclockwise and clockwise.

[0035] As shown in FIGS. **3** and **4**, each cam first slot of the plurality of cam first slots **62** extends from the cam first side **56** to the cam second side **58** and has a cam first slot first end **68**, a cam first

slot second end **70**, a cam first slot first portion **72**, and a cam first slot second portion **74**. The cam first slot first end **68** is disposed a first distance **69** from the tool body central axis **13**. The cam first slot second end **70** is disposed a second distance **71** from the tool body central axis **13** that is different than the first distance **69**. In the illustrated embodiment, the second distance **71** is greater than the first distance **69**. As shown in FIG. 4A, the cam first slot first portion **72** is disposed at an angle **73** relative to the cam first slot second portion **74** that is less than 180 degrees. Stated otherwise, the cam first portion **72** curves in a first direction and the cam second portion **74** curves in a second different direction, that is different (e.g., opposite) than the first direction). In the illustrated embodiment, the first direction is radially inward and the second direction is radially outward. This structural arrangement defines a rollover point **76** having an apex **77** within each cam first slot of the plurality of cam first slots **62**, which a respective carrier rod of the plurality of carrier rods **16** interacts with to lock the carrier rod and a respective insert of the plurality of indexable cutting inserts **18** when the cam **14** is in the cam first position. For example, as a respective carrier rod of the plurality of carrier rods **16** moves within its cam first slot of the plurality of cam first slots **62** from its carrier rod first position to its carrier rod second position, as described in more detail herein, it travels over the apex **77** and is maintained in the carrier rod second position until a force is applied to the cam **14** to move it to the carrier rod first position.

[0036] As shown in FIGS. 3 and 4, each cam second slot of the plurality of cam second slots **64** is disposed between a respective cam first slot of the plurality of cam first slots **62** and the cam periphery **60**. Each cam second slot of the plurality of cam second slots **64** extends from the cam first side **56** to the cam second side **58** and has a cam second slot first end **80** and a cam second slot second end **82**. The cam second slot first end **80** is disposed a first distance **81** from the tool body central axis **13**. The cam second slot second end **82** is disposed a second distance **83** from the tool body central axis **13** that is different than the first distance **81**. In the illustrated embodiment, the second distance **83** is greater than the first distance **81**. The inclusion of a cam second slot positioned adjacent to a respective cam first slot provides flexibility to the cam first slot that can account for different tolerances between inserts, insert pockets, and other components within the system and also result in fully seating each insert against an insert pocket tangential wall.

[0037] While the cam **14** has been illustrated as having a particular structural arrangement and as including a plurality of cam first slots **62**, a plurality of cam second slots **64**, and a cam passageway **66**, a cam can include any suitable structural arrangement and number of features. Selection of a suitable structural arrangement for a cam and of the type and number of features to include on a cam can be based on various considerations, including the intended use of the indexing system of which the cam is a component. Examples of numbers of cam first slots and cam second slots considered suitable to include on a cam include one, at least one, two, a plurality, three, four, five, six, seven, eight, nine, ten, more than ten, and any other number considered suitable for a particular embodiment. In the illustrated embodiment, the plurality of cam first slots **62** includes ten slots and the plurality of cam second slots **64** includes ten slots. Examples of suitable structural arrangements for a cam include forming a cam as a disk, bar, or tubular member. Examples of suitable structural arrangements for a cam passageway include any structural arrangement configured to receive a portion of a tool, such as those that have cross-sectional shapes that are hexagonal and can receive a hex tool, those that have square cross-sectional shapes that are square and can receive a square-head tool, and any other shape considered suitable for a particular embodiment.

[0038] Each carrier rod of the plurality of carrier rods **16** is partially disposed with a respective bore of the plurality of bores **38**, is partially disposed within a respective slot of the plurality of slots **40**, is partially disposed within a respective cam first slot of the plurality of cam first slots **62**, and contacts the cam **14**. Each carrier rod of the plurality of carrier rods **16** is moveable between a carrier rod first position, as shown in FIGS. 1, 4, 6, 8, 10, and 13, and a carrier rod second position, as shown in FIGS. 2, 5, 7, and 9. Each carrier rod of the plurality of carrier rods **16** is in the carrier rod first position when the cam **14** is in the cam first position and is in the carrier rod second

position when the cam **14** is in the cam second position.

[0039] In the illustrated embodiment, and as shown in FIGS. **3** and **11**, each carrier rod of the plurality of carrier rods **16** has a carrier rod lengthwise axis **89**, a carrier rod shaft **90**, and a carrier rod pin **92** that extends from the carrier rod shaft **90**. The carrier rod shaft **90** has a shaft first end **94**, a shaft second end **96**, a shaft first passageway **98**, and a shaft second passageway **100**. The shaft first end **94** is disposed within the tool body central passageway **30**. The shaft second end **96** is attached to a respective indexable cutting insert of the plurality of indexable cutting inserts **18** using an insert screw **102**, as described in more detail herein. The shaft first passageway **98** is sized and configured to receive a portion of the carrier rod pin **92**. The shaft second passageway **100** is sized and configured to receive a portion of an insert screw **102**, as described in more detail herein.

[0040] The carrier rod shaft **90** is moveable within its respective bore of the plurality of bores **38** when the cam **14** rotates between the cam first position and the cam second position. In the illustrated embodiment, rotational movement of the cam **14** results in linear movement of the carrier rod shaft **90**. As shown in FIGS. **4** and **5**, the shaft second end **96** is disposed a shaft first distance **91** from the tool body central axis **13** when each carrier rod of the plurality of carrier rods **16** is in the carrier rod first position and a shaft second distance **93** from the tool body central axis **13** when each carrier rod of the plurality of carrier rods **16** is in the carrier rod second position. The shaft second distance **93** is different than the shaft first distance **91**. In the illustrated embodiment, the shaft second distance **93** is greater than the shaft first distance **91**.

[0041] The carrier rod pin **92** is disposed within the shaft first passageway **98** and at an angle **95** relative to the carrier rod shaft **90** that is less than 90 degrees. In the illustrated embodiment, the carrier rod pin **92** is releasably attached to the carrier rod shaft **90** using a threaded connection. However, in alternative embodiments a carrier rod pin can be permanently attached to a carrier rod shaft or attached using other techniques or methods of attachment. Furthermore, in alternative embodiments a carrier rod pin can be disposed at any suitable angle relative to a carrier rod shaft, such as those that are greater than, less than, or equal to 90 degrees. The carrier rod pin **92** of each carrier rod of the plurality of carrier rods **16** is disposed within a respective cam first slot of the plurality of cam first slots **62** and is moveable within its respective cam first slot when the cam **14** rotates between the cam first position and the cam second position. As shown in FIGS. **4** and **5**, the carrier rod pin **92** of each carrier rod of the plurality of carrier rods **16** is disposed a pin first distance **97** from the tool body central axis **13** when each carrier rod of the plurality of carrier rods **16** is in the carrier rod first position and a pin second distance **99** from the tool body central axis **13** when each carrier rod of the plurality of carrier rods **16** is in the carrier rod second position. The pin second distance **99** is different than the pin first distance **97**. In the illustrated embodiment, the pin second distance **99** is greater than the pin first distance **97**.

[0042] While a plurality of carrier rods **16** has been illustrated as being included in the indexing system **10** and each carrier rod of the plurality of carrier rods **16** has been illustrated as having a particular structural arrangement, an indexing system can include any suitable number of carrier rods having any suitable structural arrangement. Selection of a suitable structural arrangement for a carrier rod and of a suitable number of carrier rods to include in an indexing system can be based on various considerations, including the intended use of the indexing system of which a carrier rod is a component. Examples of numbers of carrier rods considered suitable to include in an indexing system include one, at least one, two, a plurality, three, four, five, six, seven, eight, nine, ten, more than ten, and any other number considered suitable for a particular embodiment. In the illustrated embodiment, the plurality of carrier rods **16** includes ten carrier rods.

[0043] Each indexable cutting insert of the plurality of indexable cutting inserts **18** is attached to a respective carrier rod of the plurality of carrier rods **16** using an insert screw **102** and is moveable between a clamped position, as shown in FIGS. **1**, **4**, **6**, **8**, **10**, and **13**, and an indexing position, as shown in FIGS. **2**, **5**, **7**, and **9**. While an insert screw **102** has been illustrated as attaching a indexable cutting insert to a carrier rod any suitable technique or method of attachment can be

utilized to accomplish attachment between an indexable cutting insert and a carrier rod.

[0044] Each indexable cutting insert of the plurality of indexable cutting inserts **18** is in the clamped position when a respective carrier rod of the plurality of carrier rods **16** is in the carrier rod first position and the cam **14** is in the cam first position. Each indexable cutting insert of the plurality of indexable cutting inserts **18** is in the indexing position when a respective carrier rod of the plurality of carrier rods **16** is in the carrier rod second position and the cam **14** is in the cam second position. Each indexable cutting insert of the plurality of indexable cutting inserts **18** contacts an insert pocket tangential wall **52** of a respective insert pocket of the plurality of insert pockets **42** in the clamped position. Each indexable cutting insert of the plurality of indexable cutting inserts **18** does not contact an insert pocket tangential wall **52** of a respective insert pocket of the plurality of insert pockets **42** in the indexing position.

[0045] As shown in FIGS. **6**, **7**, **8**, and **9**, each indexable cutting insert of the plurality of indexable cutting inserts **18** is a double-sided insert and has an insert center axis **103**, an insert first side **104**, an insert second side **106**, an insert passageway **108**, a plurality of side support surfaces **110**, and a plurality of insert recesses **112**. The insert passageway **108** extends from the insert first side **104** to the insert second side **106** and is sized and configured to receive the insert screw **102** to accomplish attachment of an indexable cutting insert of the plurality of indexable cutting inserts **18** to a respective carrier rod of the plurality of carrier rods **16**. An insert screw can have any suitable structural arrangement to accomplish attachment between an insert and a carrier rod. For example, an insert screw can have a threaded portion along its shaft that has an outside diameter that is less than the inside diameter of an insert passageway. Alternatively, an insert screw can have a shaft that is partially threaded to allow for attachment to a carrier rod and rotation of an insert on the insert screw along the unthreaded portion. The plurality of side support surfaces **110** is disposed around the insert center axis **103** and is arranged to be supported by the plurality of side contact surfaces **54** of the tool body **12**. A first set of recesses **114** of the plurality of insert recesses **112** extends from the insert first side **104** toward the insert second side **106**. Each recess in the first set of recesses **114** is sized and configured to receive a projection of the plurality of projections **132** of a shim of the plurality of shims **20**. A second set of recesses **116** of the plurality of insert recesses **112** extends from the insert second side **106** toward the insert first side **104**. Each recess in the second set of recesses **116** is sized and configured to receive a projection of the plurality of projections **132** of a shim of the plurality of shims **20**. Alternative embodiments can omit a plurality of recesses and define other structure to accomplish a mating arrangement between an indexable cutting insert and a shim. For example, an insert can comprise a single sided positive insert or a double sided negative insert with no shim.

[0046] Each indexable cutting insert of the plurality of indexable cutting inserts **18** is rotatable around a respective insert center axis **103** and is free to turn in both a clockwise direction and counterclockwise direction around the insert center axis **103** when in the indexing position. In the embodiment shown, each indexable cutting insert of the plurality of indexable cutting inserts **18** is hexagonal and has a rotational symmetry around the insert center axis **103**. However, in alternative embodiments, an indexable cutting insert included in an indexing system can have any suitable number of side support surfaces, cutting surfaces, and have any suitable shape.

[0047] As shown in FIGS. **1** and **2**, the insert center axis **103** of each indexable cutting insert of the plurality of indexable cutting inserts **18** is disposed an insert first distance **105** from the tool body central axis **13** when in the clamped position and an insert second distance **107** from the tool body central axis **13** when in the indexing position. The insert second distance **107** is different than the insert first distance **105**. In the illustrated embodiment, the insert second distance **107** is greater than the insert first distance **105**. As shown in FIG. **7**, each indexable cutting insert of the plurality of indexable cutting inserts **18** is disposed about 7 mm from a first side contact surface **118** of the plurality of side contact surfaces **54** when in the indexing position. In the clamped position, as shown in FIG. **6**, a first side support surface **120** of the plurality of side support surfaces **110**

contacts a first side contact surface **118** of the plurality of side contact surfaces **54** and a second side support surface **122** of the plurality of side support surfaces **110** contacts a second side contact surface **124** of the plurality of side contact surfaces **54**. In the indexing position, as shown in FIG. 7, the first side support surface **120** of the plurality of side support surfaces **110** does not contact the first side contact surface **118** of the plurality of side contact surfaces **54** and the second side support surface **122** of the plurality of side support surfaces **110** does not contact the second side contact surface **124** of the plurality of side contact surfaces **54**.

[0048] While a plurality of indexable cutting inserts **18** has been illustrated as being included in the indexing system **10** and each indexable cutting insert of the plurality of indexable cutting inserts **18** has been illustrated as having a particular structural arrangement, an indexing system can include any suitable number of indexable cutting inserts having any suitable structural arrangement. Selection of a suitable structural arrangement for an indexable cutting insert and of a suitable number of indexable cutting inserts to include in an indexing system can be based on various considerations, including the intended use of the indexing system of which an indexable cutting inserts is a component. Examples of numbers of indexable cutting inserts considered suitable to include in an indexing system include one, at least one, two, a plurality, three, four, five, six, seven, eight, nine, ten, more than ten, and any other number considered suitable for a particular embodiment. In the illustrated embodiment, the plurality of indexable cutting inserts **18** includes ten cutting inserts. Furthermore, while each indexable cutting insert of the plurality of indexable cutting inserts **18** has been illustrated as disposed about 7 mm from a first side contact surface **118** of the plurality of side contact surfaces **54** when in the indexing position, an indexable cutting insert can be disposed any suitable distance relative to a first side contact surface when in an indexing position. For example, the linear distance an indexable cutting insert is disposed from a first side contact surface when in the indexing position is dependent upon the size of the indexable cutting insert and/or the geometry of an insert pocket.

[0049] Each shim of the plurality of shims **20** is disposed between a respective indexable cutting insert of the plurality of indexable cutting inserts **18** and the tool body **12**. As shown in FIG. 12, each shim of the plurality of shims **20** has a shim first side **126**, a shim second side **128**, a shim passageway **130**, and a plurality of shim projections **132**. The shim passageway **130** extends from the shim first side **126** to the shim second side **128** and is sized and configured to receive a portion of the insert screw **102** to accomplish attachment of a shim to a carrier rod of the plurality of carrier rods **16**. Each projection of the plurality of shim projections **132** extends from the shim first side **126** and away from the shim second side **128** and is sized and configured to be received by a respective recess of the plurality of insert recesses **112**. The mating configuration between a shim of the plurality of shims **20** and a respective indexable cutting insert of the plurality of indexable cutting inserts **18** provides a mechanism for preventing a positive cutting edge of an indexable cutting insert from catching on the insert pocket tangential wall **52** when the indexable cutting insert is moved between a clamped position and an indexing position. Alternative embodiments, however, can omit a plurality of projections and define other structure to accomplish a mating configuration between a shim and an indexable cutting insert.

[0050] While a plurality of shims **20** has been illustrated as being included in the indexing system **10** and each shim of the plurality of shims **20** has been illustrated as having a particular structural arrangement, an indexing system can include any suitable number of shims having any suitable structural arrangement. Selection of a suitable structural arrangement for a shim and of a suitable number of shims to include in an indexing system can be based on various considerations, including the intended use of the indexing system of which a shim is a component. Examples of numbers of shims considered suitable to include in an indexing system include one, at least one, two, a plurality, three, four, five, six, seven, eight, nine, ten, more than ten, and any other number considered suitable for a particular embodiment. In the illustrated embodiment, the plurality of shims **20** includes ten shims.

[0051] The retaining ring **22** is disposed within the tool body recess **32** and releasably attaches the cam **14** to the tool body **12**. As shown in FIGS. **1** and **3**, the retaining ring **22** has a retaining ring first end **134**, a retaining ring second end **136**, and is moveable between a first configuration and a second configuration. In the first configuration, the retaining ring **22** has a first outside diameter **133** that is less than the inside diameter **135** of the tool body central passageway **30** at the tool body recess **32**. In the second configuration, as shown in FIG. **13**, the retaining ring **22** has a second outside diameter **137** that is greater than the inside diameter **135** of the tool body central passageway **30** at the tool body recess **32**.

[0052] As shown in FIG. **13**, in an example embodiment, the indexing system **10** can be attached to a shell mill adapter, or other component, using a locking bolt **44**. The locking bolt **44** has a locking bolt outside diameter **141** that is less than the inside diameter **67** of the cam passageway **66** and greater than the smallest inside diameter **143** of the tool body central passageway **30**. To position the locking bolt **44**, the locking bolt **44** is passed through the cam passageway **66** and into the tool body central passageway **30**. Subsequently, the locking bolt **44** is attached to a shell mill adapter or other component (e.g., using a threaded connection). The structural arrangement of the indexing system **10** allows for attachment of the indexing system **10** to a shell mill adaptor or other component without having to disassemble the indexing system **10**. A tool, such as a hex tool, can then be positioned within the cam passageway **66** and rotated clockwise or counterclockwise to move the cam **14** between the cam first position and the cam second position. The indexing systems described herein provide for increased productivity as a result of all indexable cutting inserts being indexable simultaneously and without removal from a tool body.

[0053] FIGS. **14** and **15** illustrate a second example indexing system **210**. The indexing system **210** has a tool body **212**, a cam **214**, a plurality of carrier rods **216**, a plurality of indexable cutting inserts **218**, and a plurality of ball bearings **222**. In the embodiment shown, each indexable cutting insert of the plurality of indexable cutting inserts **218** are single-sided inserts, which do not require a shim.

[0054] In the illustrated embodiment, the tool body first shoulder **234** is disposed between the tool body first end **224** and the tool body second end **226** and provides a mechanical stop to advancement of a locking bolt **244**, as shown in FIG. **15** and described in more detail herein. The cam **214** is attached to the tool body **212**, is disposed within the tool body central passageway **230**, and is rotatable around the tool body central axis **213**. In the illustrated embodiment, the cam **214** has a cam first side **256**, a cam second side **258**, a cam periphery **260**, a plurality of cam first slots **262**, and a cam passageway **266** that extends from the cam first side **256** to the cam second side **258**.

[0055] Each insert pocket of the plurality of insert pockets **242** has an insert pocket terminal end **250**, an insert pocket tangential wall **252**, and a plurality of side contact surfaces **254**. In the embodiment shown, the bore lengthwise axis **245** of each bore of the plurality of bores **238** is disposed at an angle **251** relative to a hypothetical plane **253** that is disposed parallel to a respective insert pocket tangential wall **252** of an insert pocket of the plurality of insert pockets **242**. In the illustrated embodiment, the angle **251** is less than 5 degrees.

[0056] Each carrier rod of the plurality of carrier rods **216** is partially disposed with a respective bore of the plurality of bores **238**, is partially disposed within a respective slot of the plurality of slots **240**, is partially disposed within a respective cam first slot of the plurality of cam first slots **262**, and contacts the cam **214**. In the embodiment shown, and as shown in FIG. **16**, each carrier rod shaft **290** has a shaft first end **294**, a shaft second end **296**, a shaft first passageway **298**, a shaft second passageway **300**, and a chamber **302**.

[0057] The chamber **302** extends from the shaft first end **294**, toward the shaft second end **296**, and has a chamber first portion **304** and a chamber second portion **306**. The chamber first portion **304** extends from the shaft first end **294**, toward the shaft second end **296**, through the shaft first passageway **298**, and has a first inside diameter **305**. The chamber second portion **306** extends from

the chamber first portion **304**, toward the shaft second end **296**, and has a second inside diameter **307** that is greater than the first inside diameter **305**. In the embodiment shown, the chamber first portion **304** is elongated, the chamber second portion **306** is bulbous (e.g., revolved elliptical shape), and the carrier rod shaft **290** is manufactured using 3D printing. However, alternative embodiments can include a chamber having any suitable structural arrangement and can include a carrier rod shaft manufactured using any suitable method or technique to achieve a reduction of the stiffness in the system. The inclusion of a chamber **302** reduces stiffness relative to carrier rods that omit a chamber and allows a shaft to account for differences in tolerances between insert pockets. A carrier rod, such as a carrier rod of the plurality of carrier rods **216**, can be used as an alternative to the inclusion of a cam second slot or in combination with the inclusion of a cam second slot depending on the stiffness needed in a system.

[0058] The plurality of ball bearings **222** is disposed within the tool body central passageway **230** and between the tool body **212** and the cam **214**. The plurality of ball bearings **222** are disposed between an inner race **270** and an outer race **272**. The structural arrangement between the tool body **212**, the cam **214**, the inner race **270**, the outer race **272**, and the plurality of ball bearings **222** provide a mechanism for both attaching the cam **214** to the tool body **212** and allowing for rotational movement of the cam **214** relative to a tool body **212** during use. An example assembly includes pressing the cam **214** into the inner race **270** and then pressing the outer race **272** into the tool body passageway **230**. Alternatively, an inner race can be defined by a cam periphery and an outer race can be defined by a tool body.

[0059] As shown in FIG. **15**, in an example embodiment, the indexing system **210** can be attached to a shell mill adapter, or other component, using a locking bolt **244**. The locking bolt **244** has an outside diameter **341** that is less than the inside diameter **267** of the cam passageway **266** and greater than the smallest inside diameter **343** of the tool body central passageway **230**. To position the locking bolt **244**, the cam **214** does not need to be removed from the tool body **212**. Rather, the locking bolt **244** is passed through the cam passageway **266**, contacts the tool body first shoulder **234**, and is attached to a shell mill adapter or other component (e.g., using a threaded connection). The indexing system **210** is then ready to be used.

[0060] FIG. **17** illustrates an alternative carrier rod shaft **490** that can be included in an indexing system. The carrier rod shaft **490** has a shaft first end **494**, a shaft second end **496**, a shaft first passageway **498**, a shaft second passageway **500**, and a shaft third passageway **502**. The shaft third passageway **502** is disposed between the shaft first passageway **498** and the shaft second passageway **500** and has an elliptical cross-sectional shape. However, in alternative embodiments, a shaft third passageway can have any suitable shape. The inclusion of a shaft third passageway **502** reduces stiffness relative to carrier rods that omit a third passageway and allows a shaft to account for differences in tolerances between insert pockets.

[0061] Those with ordinary skill in the art will appreciate that various modifications and alternatives for the described and illustrated embodiments can be developed in light of the overall teachings of the disclosure, and that the various elements and features of one example described and illustrated herein can be combined with various elements and features of another example without departing from the scope of the invention. Accordingly, the particular examples disclosed herein have been selected by the inventor(s) simply to describe and illustrate examples of the invention and are not intended to limit the scope of the invention or its protection, which is to be given the full breadth of the appended claims and any and all equivalents thereof.

Claims

1. An indexing system comprising: a tool body rotatable around a tool body central axis, the tool body having a tool body central passageway, a bore, and an insert pocket having a plurality of side contact surfaces; a cam attached to the tool body, disposed within the tool body central passageway,

and rotatable around the tool body central axis between a cam first position and a cam second position; a carrier rod partially disposed within the bore, contacting the cam, and moveable between a carrier rod first position and a carrier rod second position, the carrier rod is in the carrier rod first position when the cam is in the cam first position, the carrier rod is in the carrier rod second position when the cam is in the cam second position; and an indexable cutting insert attached to the carrier rod, the indexable cutting insert having an insert center axis and a plurality of side support surfaces, the indexable cutting insert moveable between a clamped position and an indexing position, the indexable cutting insert is in the clamped position when the carrier rod is in the carrier rod first position, the indexable cutting insert is in the indexing position when the carrier rod is in the carrier rod second position, in the clamped position a first side support surface of the plurality of side support surfaces contacts a first side contact surface of the plurality of side contact surfaces and a second side support surface of the plurality of side support surfaces contacts a second side contact surface of the plurality of side contact surfaces, in the indexing position the first side support surface does not contact the first side contact surface, the second side support surface does not contact the second side contact surface, and the indexable cutting insert is rotatable around the insert center axis.

2. The indexing system of claim 1, wherein the cam has a cam first slot, the cam first slot having a cam first slot first end and a cam first slot second end, the cam first slot first end disposed a first distance from the tool body central axis, the cam first slot second end disposed a second distance from the tool body central axis that is different than the first distance; and wherein the carrier rod is disposed within the cam first slot.

3. The indexing system of claim 2, wherein the second distance is greater than the first distance.

4. The indexing system of claim 2, wherein the carrier rod has a carrier rod shaft and a carrier rod pin extending from the carrier rod shaft, the carrier rod pin disposed within the cam first slot, the carrier rod pin disposed a pin first distance from the tool body central axis when the carrier rod is in the carrier rod first position and a pin second distance from the tool body central axis when the carrier rod is in the carrier rod second position, the pin second distance being different than the pin first distance.

5. The indexing system of claim 4, wherein the pin second distance is greater than the pin first distance.

6. The indexing system of claim 4, wherein the carrier rod pin moves within the cam first slot when the cam rotates between the cam first position and the cam second position.

7. The indexing system of claim 4, wherein the carrier rod pin is releasably attached to the carrier rod shaft.

8. The indexing system of claim 2, wherein the cam first slot has a cam first slot first portion and a cam first slot second portion, the cam first slot first portion curving in a first direction and the cam first slot second portion curving in a second, different direction such that a rollover point having an apex is defined within the cam first slot.

9. The indexing system of claim 2, wherein the cam has a cam periphery and a cam second slot disposed between the cam first slot and the cam periphery.

10. The indexing system of claim 1, wherein the insert pocket has an insert pocket terminal end and an insert pocket tangential wall; wherein the tool body has a slot extending from the insert pocket terminal end toward the tool body central passageway and into the insert pocket tangential wall; and wherein the carrier rod is partially disposed within the slot.

11. The indexing system of claim 10, wherein the bore extends from a bore first end disposed at the tool body central passageway to a bore second end disposed at the slot.

12. The indexing system of claim 1, wherein the insert pocket has an insert pocket tangential wall; and wherein the bore has a bore lengthwise axis that is not parallel to the insert pocket tangential wall.

13. The indexing system of claim 1, wherein the tool body has a tool body periphery; wherein the

bore extends from a bore first end disposed at the tool body central passageway toward the tool body periphery to a bore second end; wherein the insert pocket has an insert pocket tangential wall; and wherein the bore extends toward the insert pocket tangential wall from the bore first end to the bore second end.

14. The indexing system of claim 13, wherein the bore has a bore lengthwise axis disposed at an angle of about 5 degrees relative to a hypothetical plane that is parallel to the insert pocket tangential wall.

15. The indexing system of claim 1, wherein the cam has about 36 degrees of rotation around the tool body central axis.

16. The indexing system of claim 1, further comprising a shim disposed between the indexable cutting insert and the tool body.

17. The indexing system of claim 1, wherein the insert pocket has an insert pocket tangential wall; wherein the indexable cutting insert contacts the insert pocket tangential wall in the clamped position; and wherein the indexable cutting insert does not contact the insert pocket tangential wall in the indexing position.

18. The indexing system of claim 1, wherein the insert center axis is disposed an insert first distance from the tool body central axis when the indexable cutting insert is in the clamped position and an insert second distance from the tool body central axis when the indexable cutting insert is in the indexing position, the insert second distance being different than the insert first distance.

19. An indexing system comprising: a tool body rotatable around a tool body central axis, the tool body having a tool body central passageway, a bore, and an insert pocket having a plurality of side contact surfaces and an insert pocket tangential wall, the bore having a bore lengthwise axis that is not parallel to the insert pocket tangential wall; a cam attached to the tool body, disposed within the tool body central passageway, and rotatable around the tool body central axis between a cam first position and a cam second position, the cam having a cam first slot, the cam first slot having a cam first slot first end and a cam first slot second end, the cam first slot first end disposed a first distance from the tool body central axis, the cam first slot second end disposed a second distance from the tool body central axis that is different than the first distance; a carrier rod partially disposed within the bore, disposed within the cam first slot, and moveable between a carrier rod first position and a carrier rod second position, the carrier rod is in the carrier rod first position when the cam is in the cam first position, the carrier rod is in the carrier rod second position when the cam is in the cam second position; and an indexable cutting insert attached to the carrier rod, the indexable cutting insert having an insert center axis and a plurality of side support surfaces, the indexable cutting insert moveable between a clamped position and an indexing position, the indexable cutting insert is in the clamped position when the carrier rod is in the carrier rod first position, the indexable cutting insert is in the indexing position when the carrier rod is in the carrier rod second position, in the clamped position a first side support surface of the plurality of side support surfaces contacts a first side contact surface of the plurality of side contact surfaces and a second side support surface of the plurality of side support surfaces contacts a second side contact surface of the plurality of side contact surfaces, in the indexing position the first side support surface does not contact the first side contact surface, the second side support surface does not contact the second side contact surface, and the indexable cutting insert is rotatable around the insert center axis.

20. An indexing system comprising: a tool body rotatable around a tool body central axis, the tool body having a tool body central passageway, a bore, and an insert pocket having a plurality of side contact surfaces and an insert pocket tangential wall, the bore having a bore lengthwise axis that is not parallel to the insert pocket tangential wall; a cam attached to the tool body, disposed within the tool body central passageway, and rotatable around the tool body central axis between a cam first position and a cam second position, the cam having a cam periphery, a cam first slot, and a cam second slot, the cam first slot having a cam first slot first end and a cam first slot second end, the

cam first slot first end disposed a first distance from the tool body central axis, the cam first slot second end disposed a second distance from the tool body central axis that is greater than the first distance, the cam second slot disposed between the cam first slot and the cam periphery; a carrier rod partially disposed within the bore, disposed within the cam first slot, and moveable between a carrier rod first position and a carrier rod second position, the carrier rod is in the carrier rod first position when the cam is in the cam first position, the carrier rod is in the carrier rod second position when the cam is in the cam second position; and an indexable cutting insert attached to the carrier rod, the indexable cutting insert having an insert center axis and a plurality of side support surfaces, the indexable cutting insert moveable between a clamped position and an indexing position, the indexable cutting insert is in the clamped position when the carrier rod is in the carrier rod first position, the indexable cutting insert is in the indexing position when the carrier rod is in the carrier rod second position, in the clamped position a first side support surface of the plurality of side support surfaces contacts a first side contact surface of the plurality of side contact surfaces and a second side support surface of the plurality of side support surfaces contacts a second side contact surface of the plurality of side contact surfaces, in the indexing position the first side support surface does not contact the first side contact surface, the second side support surface does not contact the second side contact surface, and the indexable cutting insert is rotatable around the insert center axis.
